

Model Review: Evaluation, Diagnostics & Comparison

Task ID: TB-L11

Date: 05-Sep

Focus: KNN vs. Decision Tree Benchmarking

COMPLETED

1. PERFORMANCE BENCHMARK TABLE

Evaluation Metric	KNN (k = 8)	Decision Tree	Better Model
Overall Test Accuracy	81.50%	80.00%	KNN (+1.5%)
Precision (Class 1 - Approved)	78.95%	58.00%	KNN (High Precision)
Recall (Class 1 - Approved)	31.25%	60.42%	Decision Tree (+29.17%)
F1-Score (Class 1 - Approved)	44.78%	59.18%	Decision Tree (+14.4%)
Type I Error (False Approvals)	4 applicants	21 applicants	KNN (Lower Risk)
Type II Error (False Rejections)	33 applicants	19 applicants	Decision Tree (Higher Capture)

2. CONFUSION MATRIX DIAGNOSTICS (N = 200)

KNN (k = 8)			Decision Tree		
Actual \ Pred	Pred: 0 (Reject)	Pred: 1 (Approve)	Actual \ Pred	Pred: 0 (Reject)	Pred: 1 (Approve)
Actual: 0	TN = 148	FP = 4	Actual: 0	TN = 131	FP = 21
Actual: 1	FN = 33	TP = 15	Actual: 1	FN = 19	TP = 29

• **Characteristics:** Highly conservative; only approves when confidence is very high. Minimizes false approvals.

• **Characteristics:** Higher capture rate (Recall 60.42%); approves nearly double the eligible borrowers at the cost of higher FP.

3. BUSINESS TRADE-OFFS & STRATEGIC GUIDANCE

♥ Default Prevention Strategy (KNN)

Recommended when the financial institution prioritizes **loss avoidance and low risk**. With only **4 false positives**, it shields capital from default-prone applicants.

📈 Growth & Revenue Strategy (Decision Tree)

Recommended when targeting **portfolio expansion and interest revenue**. Captures **29 approvals (Recall: 60.42%)**, nearly double KNN's volume.

Regulatory & Explainability Advantage: The **Decision Tree** model offers transparent decision pathways (e.g., Credit Score and Income branch splits), simplifying compliance and adverse-action reporting for denied applicants.

4. REVIEW DECISION & INTEGRATION

Dual-Model Integration: Both models have been integrated into `GUI_App.py`. Underwriting officers can toggle between the **Conservative / Low-Risk Engine (KNN)** and the **Growth / Explainable Engine (Decision Tree)** depending on credit risk appetite.